NNN-C-440

March 1, 1963

SUPERSEDING
Fed. Spec. GG-C-411
November 23, 1951

FEDERAL SPECIFICATION

CLAMPS, UTILITY, LABORATORY, AND HOLDERS, BURET AND CLAMP

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers clamps and clamp holders for use in securing laboratory apparatus to support stands.

1.2 Classification.

1.2.1 *Types, classes, and sizes.* The clamps and clamp holders covered by this specification shall be of the following types, classes, and sizes, as specified (see 6.2):

Type I—Clamp holders.

Class 1—Fixed jaw.

Class 2—Swivel jaw.

Class 3—All position jaw.

Type II—Two-pronged clamps.

Class 1—Fixed, with holder.

Size 1—Small, clamp expands to 1-1/2 inches (in.).

Size 2—Large, clamp expands to 2-1/2 in.

Class 2—Swivel, with holder.

Size 1—Small, clamp expands to 1-1/2 in.

Size 2—Large, clamp expands to 2-1/2 in.

Class 3—Swivel, duplex.

Type III—Extension clamps.

Class 1—Two-pronged.

Size 1—Small, clamp expands to 1-1/2 in.

Size 2—Large, clamp expands to 2-1/2 in.

Class 2—Three-pronged.

Size 1—Small, clamp expands to 1 in.

Size 2—Large, clamp expands to 3-7/8 in.

Class 3—Universal.

Size 1—Small, clamp expands to 2 in.

Size 2—Large, clamp expands to 3 in.

Type IV—Thermometer clamp, swivel, with holder.

Type V—Clamp, double buret holder.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-A-367—Aluminum-Alloy Forgings, Heat-Treated.

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QQ-A-591—Aluminum-Alloy Die Castings.

QQ-C-523—Copper Alloy Ingots; Brass (Yellow, High Strength) Manganese and Manganese-Aluminum Bronze.

QQ-C-525—Copper Alloy Ingots; (Leaded and Nonleaded Tin Bronze, Red Brass, and Semi-Red Brass).

QQ-P-330—Phosphor Bronze Bars, Plates, Rods, Sheets, Strips, Flat Wire, and Structural and Special Shaped Sections.

QQ-S-763—Steel Bars, Shapes, and Forgings—Corrosion Resisting.

QQ-Z-363—Zinc-Base Alloy; Die Castings.

UU-T-111—Tape; Paper, Gummed (Sealing and Securing).

PPP-B-566—Boxes, Folding, Paperboard.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

PPP-B-676—Boxes, Set-Up, Paperboard.

PPP-C-843—Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-P-20689—Plastic, Plastisol: Molding, Extruding, Casting, and Dipping Compound.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-417—Rubber Compositions, Vulcanized, General Purpose, Solid (Symbols and Tests).

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

National Bureau of Standards Handbook:

H28—Screw Thread Standards for Federal Services.

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(Applications for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Base metal.* Clamps and clamp holders shall be manufactured from aluminum-base alloy or zinc-base alloy. Aluminum-base alloy metal shall conform to alloy No. A380 of QQ-A-591. Zinc-base alloy metal shall conform to composition A of QQ-Z-363.

3.1.2 *Metal fittings.* Fittings, such as screws, nuts, and rivets, for clamps and clamp holders shall be forged aluminum or chemical resistant zinc alloy for either aluminum-base alloy or zinc-base alloy clamps and clamp holders. In addition, zinc-base alloy clamps and clamp holders may have copper alloy fittings. Forged aluminum fittings shall conform in composition to 2017 of QQ-A-367. Copper alloy fittings shall conform in composition to QQ-C-523 or QQ-C-525.

3.1.3 *Springs.* Component springs of clamps shall be of phosphor bronze or corrosion-resistant steel. Phosphor bronze shall conform to composition A of QQ-P-330. Corrosion-resistant steel shall conform to class 202 or 302 of QQ-S-763.

3.1.4 *Sleeves.* Component sleeves of clamps may be of rubber, plastic, or asbestos. Unless otherwise specified by the procuring agency, rubber sleeves shall be supplied with clamps (see 6.2).

3.1.4.1 *Rubber.* Rubber for sleeves shall conform to type R, class optional, of MIL-STD-417.

3.1.4.2 *Plastic.* When plastic sleeves are specified, they shall be bonded firmly to the clamp jaw. The plastic sleeve shall be formed by hot dipping the clamp jaw in a vinyl plastisol conforming to type II of MIL-P-20689.

3.1.4.3 *Asbestos.* When asbestos sleeves are specified, they shall be non-raveling and woven from high quality asbestos yarn.

TABLE I.—*Dimensions and tolerances for clamps and clamp holders*

Item			Maximum diameter of opening, inches		Overall length, inches	± tolerance, inch	Type of jaw	
Type	Class	Size	Clamp	Holder			Clamp	Holder
I	1	—	—	11/16	5	1/4	—	V
I	2	—	—	11/16	6	1/4	—	V
I	3	—	—	11/16	5	1/8	—	V
II	1	1	1-1/2	11/16	7	1/4	Round	V-
II	1	2	2-1/2	11/16	7	1/4	Round	V-
II	2	1	1-1/2	11/16	7	1/4	Round	V-
II	2	2	2-1/2	11/16	7-1/2	1/4	Round	V-
II	3	—	1-1/2 & 2-1/2*	—	8-1/2	1/4	Round	—
III	1	1	1-1/2	—	8	1/4	Round	—
III	1	2	2-1/2	—	9	1/4	Round	—
III	2	1	1	—	6	1/4	Prong	—
III	2	2	3-7/8	—	10-1/2	1/2	Prong	—
III	3	1	2	—	8	1/4	Prong	—
III	3	2	3	—	9-1/2	1/4	Prong	—
IV	—	—	1/8 to 1/2	11/16	6	1/4	Clip	V-
V	—	—	**	11/16	10	1/4	Finger	V-

* Clamp on each end—one small, one large sized.
** Clamp shall be capable of holding mirco to 100 milliliter size bureta 5 to 20 millimeters in diameter.

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3.2 **Design.**

3.2.1 *Fitting threads.* Thumbscrews, wingnuts, and holders shall be threaded in accordance with Screw Thread Standards for Federal Services, Unified National Coarse Thread Series, Handbook H28.

3.2.2 *Dimensions.* The dimensions and tolerances shall be as shown in table I.

3.2.3 *Construction.* The illustrations shown in figure 1 are for the convenience of identification and shall not preclude clamps and clamp holders otherwise in accordance with this specification.

3.2.3.1 *Type I.* Class 1 clamp holders shall be of one-piece construction, consisting of 2 V-grooved areas at right angles to each other. Class 2 clamp holders shall consist of 2 separate V-grooved sections freely movable through an angle of 360 degrees in parallel planes and position-fixed by means of a set screw. Class 3 clamp holders shall consist of 2 V-grooved areas connected to each other by a double plate joint which keeps the V-grooved areas in planes permanently set at 90 degrees to each other. The plate joints have friction discs which allows each V-grooved area full movement through an angle of 360 degrees within each plane. The V-grooved areas of each class are fitted with thumbscrews for attachment to support rods or anchoring clamp shafts. All clamp holders shall be capable of direct attachment to rods up to 1/2 inch in diameter without the necessity of passing the clamp holders over the end of the rods. All clamp holders shall be capable of attachment to rods up to 11/16 inch in diameter by passing the clamp holder over the end of the rods.

3.2.3.2 *Type II.* Class 1 and class 2 two-pronged clamps shall consist of a V-grooved holder fitted with thumbscrews for attachment to support rods and a spring-type clamp. The clamp shall have 2 jaws that spread under spring tension and shall be adjustable to the maximum opening shown in table I by means of a single screw and wingnut. Class 1 two-pronged clamps shall be of one-piece construction. Class 2 two-pronged clamps shall be of two-piece construction capable of being extended approximately 1 inch. The clamp section shall be adjustable through 360 degrees and position-fixed by means of a thumbscrew or locknut on the extension shaft. Class 3 two-pronged clamps shall consist of 2 two-pronged clamps connected by means of a threaded shaft capable of being extended approximately 1 inch, adjustable through 360 degrees, and position-fixed by means of a locknut on the extension shaft.

3.2.3.3 *Type III.* Extension clamps shall consist of a spring-type clamp fixed to an extension arm for insertion and positioning in a type I clamp holder. Class 1 extension clamp shall have a two-pronged jaw that spreads under spring tension and be adjustable to the maximum opening shown in table I by means of a single screw and wingnut. Class 2 extension clamp shall have a two-pronged jaw and a one-pronged jaw, each independently adjustable by means of a separate thumbscrew. The one-pronged jaw shall be so located that it is equidistant from both prongs of the two-pronged jaw. Class 3 extension clamp shall be three-pronged (of similar construction to class 2) or four-pronged (2 two-pronged jaws—one jaw being smaller than the other so that the smaller jaw will pass between the prongs of the larger jaw) fitted to the extension arm by means of a threaded swivel and wingnut for positioning the clamp at any angle.

3.2.3.4 *Type IV.* Thermometer clamps shall consist of a V-grooved holder and a spring clip having a wingnut for tightening the thermometer in the clip. The holder and clip shall be connected by means of a threaded shaft capable of being extended approximately 1 inch, adjustable through 360 degrees, and position-fixed by means of a locknut on the extension shaft.

3.2.3.5 *Type V.* Clamps, double buret holder shall be constructed of a one-piece H-frame with the vertical arms curved out-

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ward. The ends of each arm shall have a suitable curved finger at right angle to the frame. Two spring-actuated gripping arms, each with a suitable curved finger at right angle to the arm, shall be attached to each side of the H-frame and aligned with the curved frame arms to form double clamps for supporting 2 burets ranging in diameter from 5 to 20 millimeters. Burets shall be firmly gripped at 2 points by the clamps thus formed without sideslipping or obscuring any of the buret graduations. Insertion, adjustment, and removal of burets shall be accomplished by applying pressure on the spring-actuated arms; each side of the buret holder shall operate separately. The buret holder shall be fitted with a setscrew, centrally located, for fastening the holder to a support rod.

3.3 **Identification marking.** Unless otherwise specified, each clamp and clamp holder shall be permanently and legibly marked with the manufacturer's name or trade name.

3.4 **Workmanship.** All clamps and clamp holders shall be free from imperfections which affect their serviceability. All parts shall be constructed of suitable gauge material to prevent deformation when used extensively in the laboratory. Edges and corners shall be finished free of burrs and fins.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1, 3.2.1, and section 5 have been satisfied.

4.3 **Lotting.** A lot of clamps or clamp holders shall consist of those clamps or clamp holders of the same type, same class, and same size made by one manufacturer and submitted for delivery at one time.

4.4 **Sampling for nondestructive examination.** Sampling shall be conducted in accordance with MIL-STD-105.

4.5 **Examination.** Samples of clamps or clamp holders and level A and level B preparation for delivery requirements shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Clamps or clamp holders.*

Categories	*Defects*
Critical:	None defined
Major:	*AQL 4.0 percent defective*
101	Dimensions not as specified
102	Fitting threads not as specified
103	Design not as specified
104	Defective (burrs, fins, or cracks)
105	Marking incorrect, illegible, or incomplete

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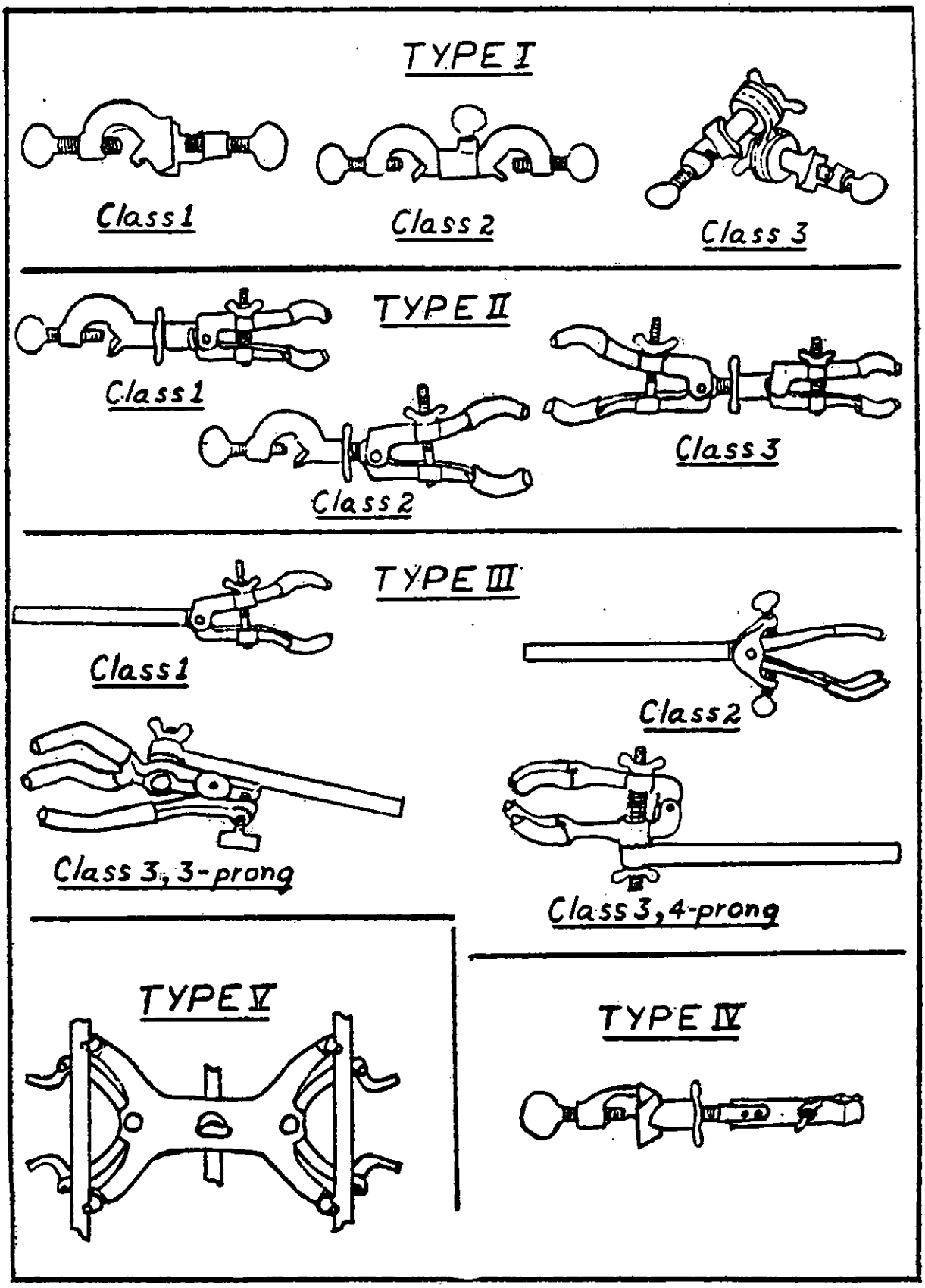


FIGURE 1. Clamps and clamp holders.

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4.5.1.2 *Level A and level B preparation for delivery (section 5).*

Categories	*Defects*
Critical:	None Defined
Major:	*AQL 2.5 percent defective*
101	Unit, intermediate, or shipping container not as specified
102	Number of clamps or holders per intermediate or shipping container not as specified or indicated
103	Unlike clamps or holders packaged per intermediate, or shipping container
104	Closure of unit, intermediate, or shipping container not as specified
105	Unit, intermediate, or shipping container damaged
106	Cushioning material not as specified
107	Marking of unit, intermediate, or shipping container illegible, incorrect, or incomplete

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* Each clamp or each clamp holder shall be packaged in a box of appropriate size conforming to PPP-B-566, style I or II, type and class optional or PPP-B-676, type, class, and style optional. The box shall be provided with fiberboard blocking or sufficient cushioning material conforming to type II, class B of PPP-C-843 to prevent shifting of contents. Box closures shall be secured with tape conforming to UU-T-111.

5.1.1.2 *Intermediate packaging.* The number of unit packages specified (see 6.2) shall be intermediately packaged in a box conforming to type I, class 1, of PPP-B-636. Each intermediate package shall contain only one type, class, and size clamp or clamp holder. Box closure shall be in accordance with the appendix of the box specification.

5.1.2 *Level C.* Clamps and clamp holders shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* The number of clamps or clamp holders specified (see 6.2) shall be packed in a box conforming to PPP-B-601, overseas type or PPP-B-621, class 2. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and contain identical quantities of like items. Gross weight shall not exceed the weight limitations of the applicable box specification or appendix thereto. Containers shall be closed and strapped in accordance with the applicable box specification and appendix thereto.

5.2.2 *Level B.* The number of clamps or clamp holders specified (see 6.2) shall be packed in shipping containers conforming to PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type I or II, class 1, style RSC, minimum bursting strength 275 pounds per square inch.

5.2.3 *Level C.* Clamps or clamp holders shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Clamps and clamp holders covered by this specification are intended for use in laboratory applications — to support or hold flasks, condensers, columns, thermometers, burets, etc.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Level of packaging and packing required.
(*c*) Type, class, and size required.
(*d*) If clamps desired with sleeves other than rubber.
(*e*) Number of unit packages per intermediate package and number of clamps or clamp holders per pack required.

6.3 Military use. Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard or military supply standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 Supersession data. The clamps and clamp holders of this specification correspond to the clamps and clamp holders of GG-C-411 as shown in table II.

6.5 Transportation description. Transportation description applicable to this item is:

Aluminum articles, not otherwise indexed by name.
Carload minimum weights 14,000 and 24,000 pounds, subject to Rule 34, Uniform Freight Classification.
Aluminum articles, not otherwise indexed.
Truckload minimum weight 14,000 pounds, subject to Rule 115, National Motor Freight Classification.
Zinc articles, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.
Zinc articles, not otherwise indexed.
Motor volume minimum weight 30,000 pounds.

TABLE II.—*Corresponding classification of clamps and clamp holders*

NNN-C-440			GG-C-411	
Type	Class	Size	Type	Size
I	1	—	7	—
I	2	—	8	—
I	3	—	—	—
II	1	1	2	Small
II	1	2	2	Large
II	2	1	6	Small
II	2	2	6	Large
II	3	—	3	—
III	1	1	4	Small
III	1	2	4	Large
III	2	1	—	—
III	2	2	5	—
III	3	1	—	—
III	3	2	—	—
IV	—	—	—	—
V	—	—	1	—

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MILITARY CUSTODIANS:

Army—MU

Navy—Md

Air Force—DAY

OTHER INTERESTS:

Army—E M O

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